

Project 2 – Digital Forensics

Use Wireshark to Analyze Packet Captures and Detect a Man-in-the-Middle (MITM) Attack via ARP Poisoning

Introduction

Digital forensics plays a critical and key role in identifying, analyzing, and documenting cyber incidents. When examining a incident, forensic analysts capture network traffic to detect malicious activities and identify potential security breaches. Among these incidents, Man-in-the-Middle (MITM) attacks are particularly significant because they allow an attacker to intercept and manipulate communication between two legitimate systems without their knowledge.

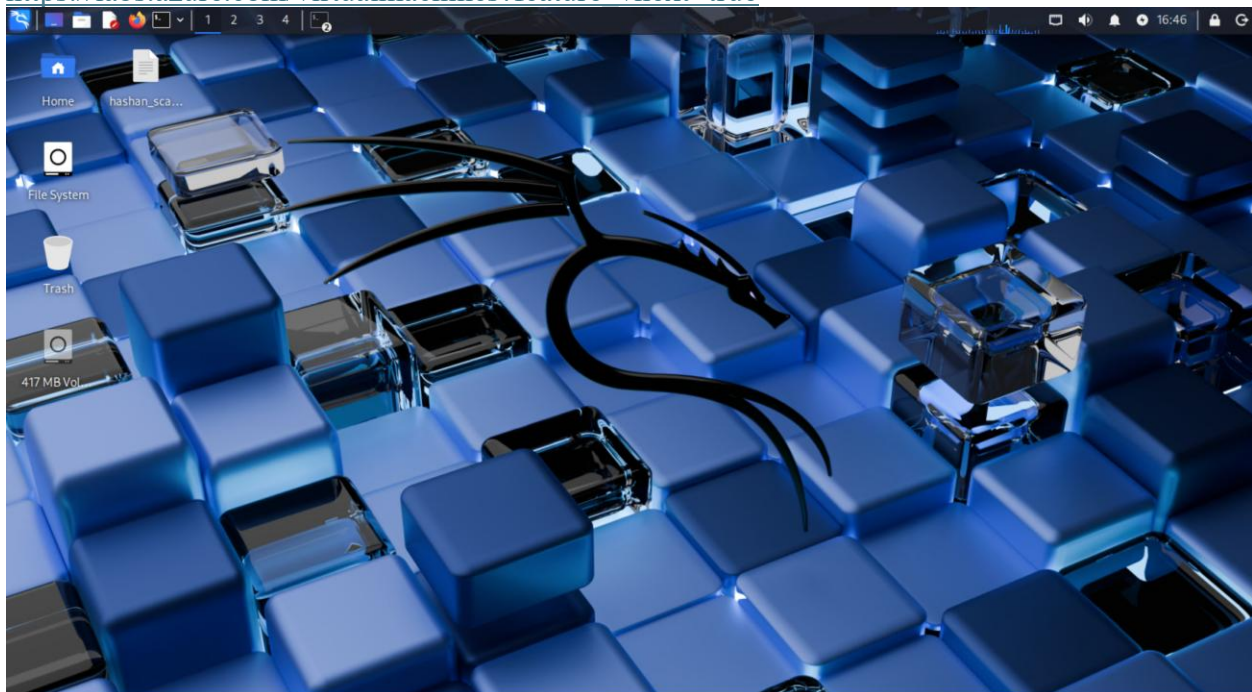
This project focuses on detecting a suspected MITM attack carried out using Address Resolution Protocol (ARP) poisoning. ARP spoofing is a cyberattack technique where the attacker manipulates the Address Resolution Protocol (ARP) to pretend to be another machine on the local network (Invicti, n.d.).

This investigation is conducted to analyze a provided packet capture *proj_part1.pcap* file using Wireshark and identify indicators of compromise associated with ARP poisoning. Before starting the investigation, verify file hash to ensure the integrity and authenticity of the evidence. Then traffic analysis conducted using endpoint statistics and detailed packet inspection. Further, checked the ARP traffic patterns and anomalies in MAC address mappings to determine whether a MITM attack has occurred.

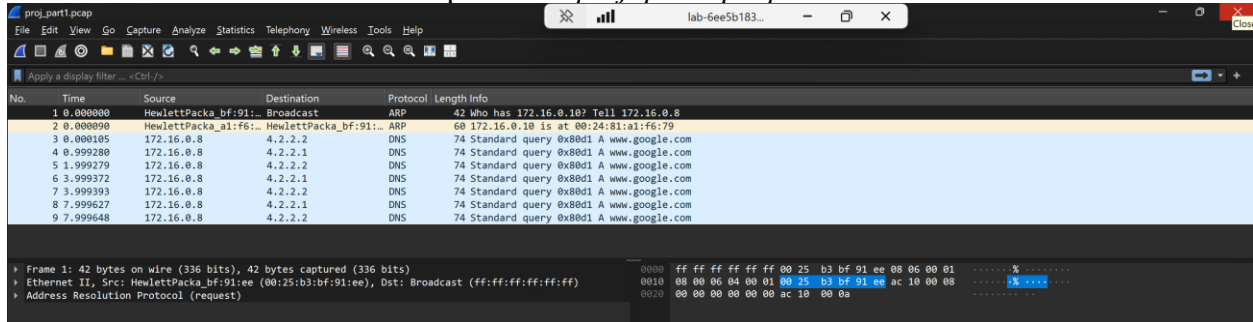
Part 1- Getting Familiar with Wireshark

1. Logged to Azure Lab environment using following web link.

https://labs.azure.com/virtualmachines?feature_vnext=true



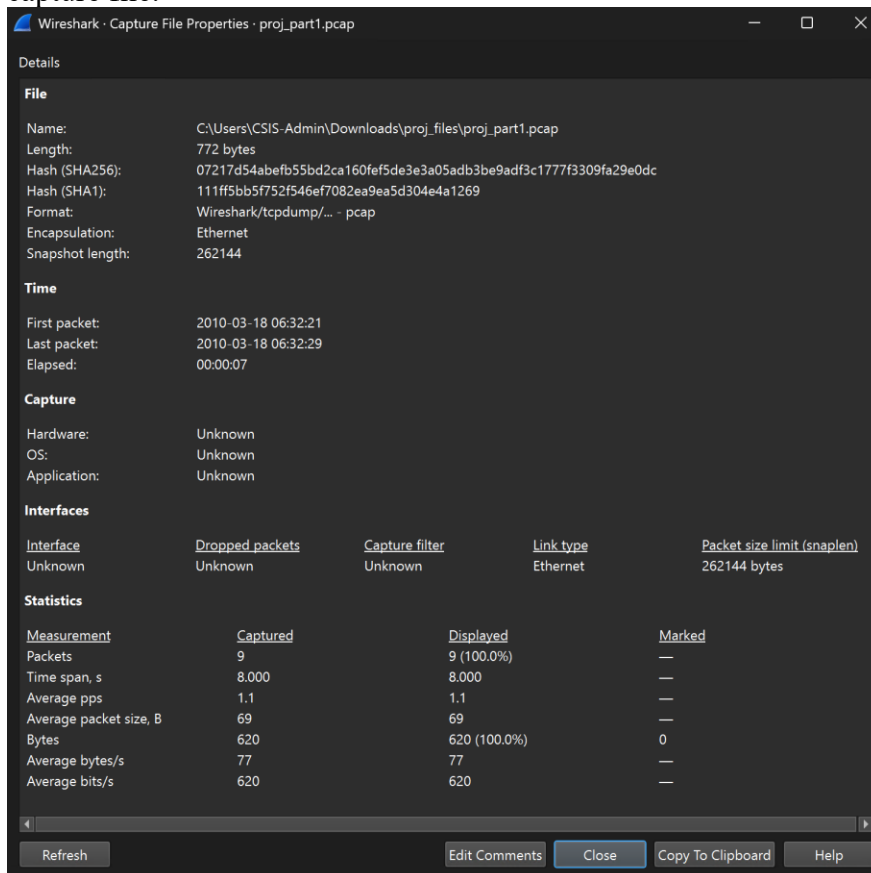
2. Launched Wireshark and opened the *proj_part1.pcap* file.



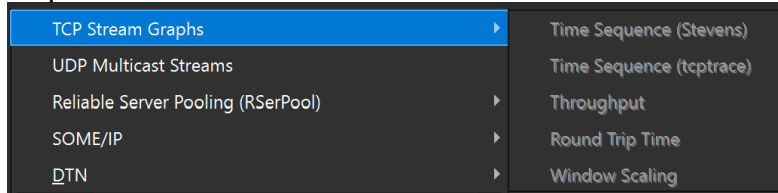
3. Checked the protocols captured from the .pcap file.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	HewlettPacka_bf:91:...	Broadcast	ARP	42	Who has 172.16.0.10? Tell 172.16.0.8
2	0.000090	HewlettPacka_a1:f6:...	HewlettPacka_bf:91:...	ARP	60	172.16.0.10 is at 00:24:81:a1:f6:79
3	0.000105	172.16.0.8	4.2.2.2	DNS	74	Standard query 0x80d1 A www.google.com
4	0.999280	172.16.0.8	4.2.2.1	DNS	74	Standard query 0x80d1 A www.google.com
5	1.999279	172.16.0.8	4.2.2.2	DNS	74	Standard query 0x80d1 A www.google.com
6	3.999372	172.16.0.8	4.2.2.1	DNS	74	Standard query 0x80d1 A www.google.com
7	3.999393	172.16.0.8	4.2.2.2	DNS	74	Standard query 0x80d1 A www.google.com
8	7.999627	172.16.0.8	4.2.2.1	DNS	74	Standard query 0x80d1 A www.google.com
9	7.999648	172.16.0.8	4.2.2.2	DNS	74	Standard query 0x80d1 A www.google.com

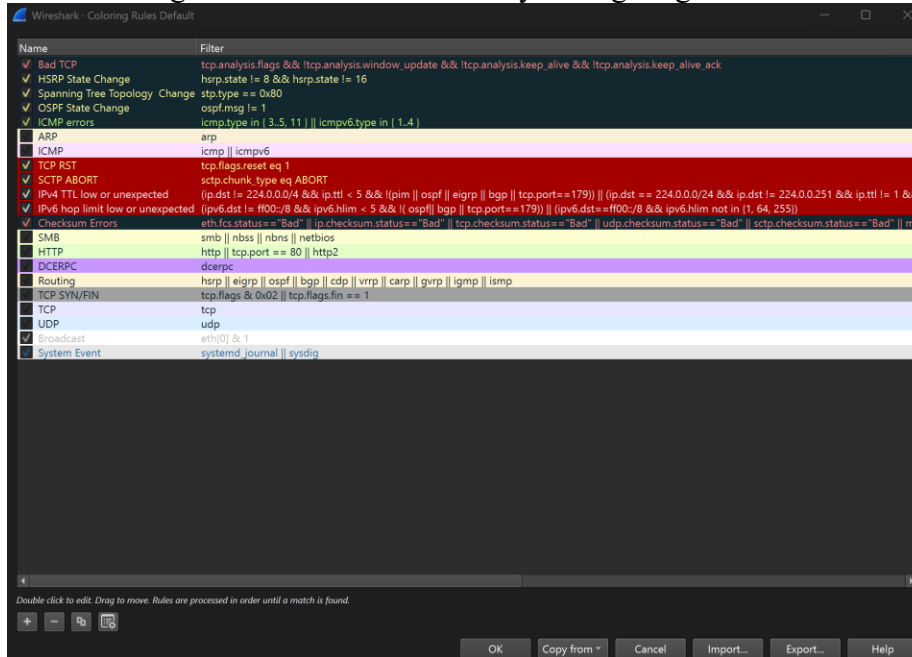
4. Then clicked Statistics on the menu > Capture File properties to examine the details of the capture file.



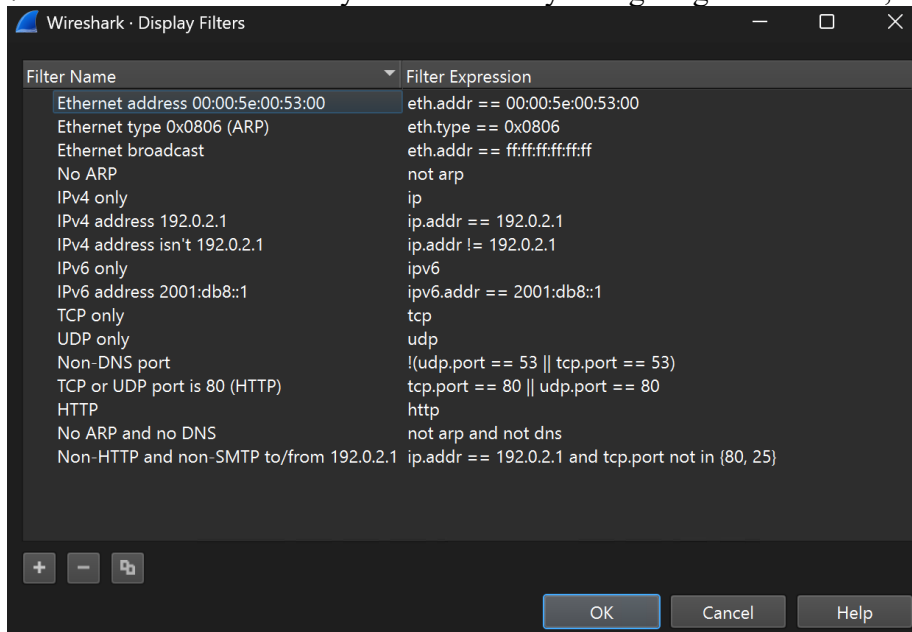
5. Tried to analyze packet flow over time by clicking Statistics > TCP Stream Graph > Time Sequence. But it does not allow me to check it.



6. Changed the color scheme by navigating to the menu View > Coloring Rules.



7. Checked the commonly used filters by navigating to the menu, Analyze > Display Filters.



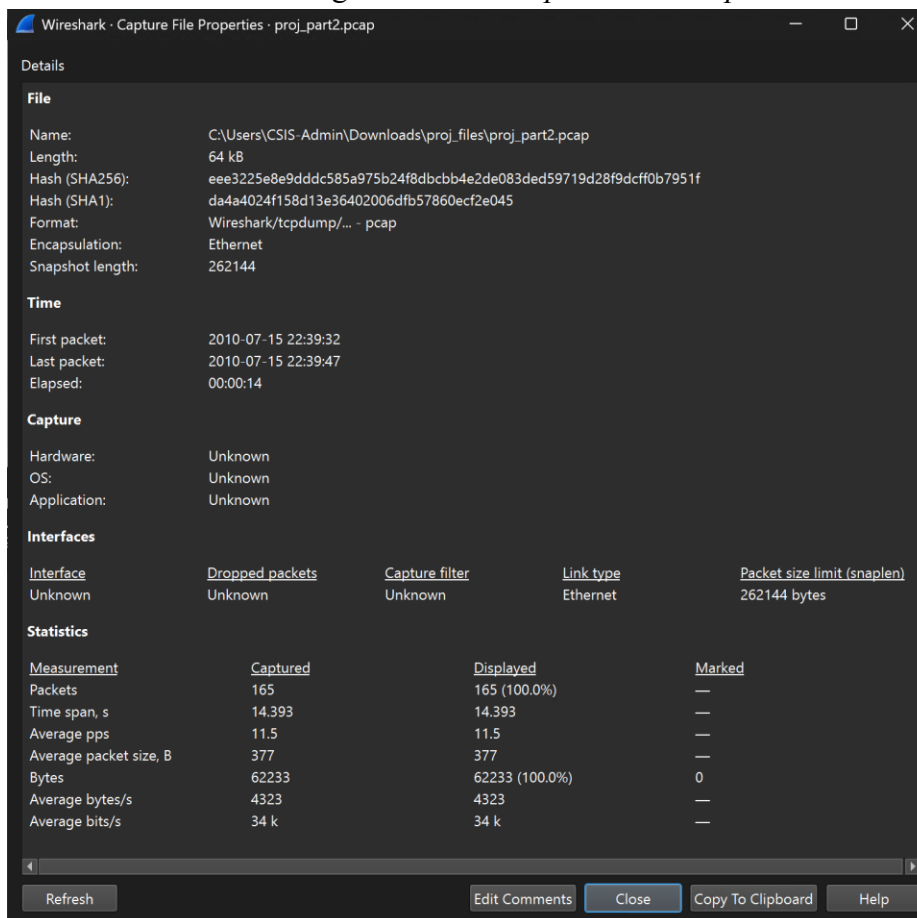
Part 2- Network Forensics Investigation, suspected Man in the Middle (MITM) attack

ARP in MITM

Address Resolution Protocol (ARP) uses two types of packets, request packets and response packets. To find a machine that has the MAC address associated with a particular IP address, the request packet is sent out as a broadcast to all the hosts in a network. The correct machine responds to the request by sending a unicast packet response.

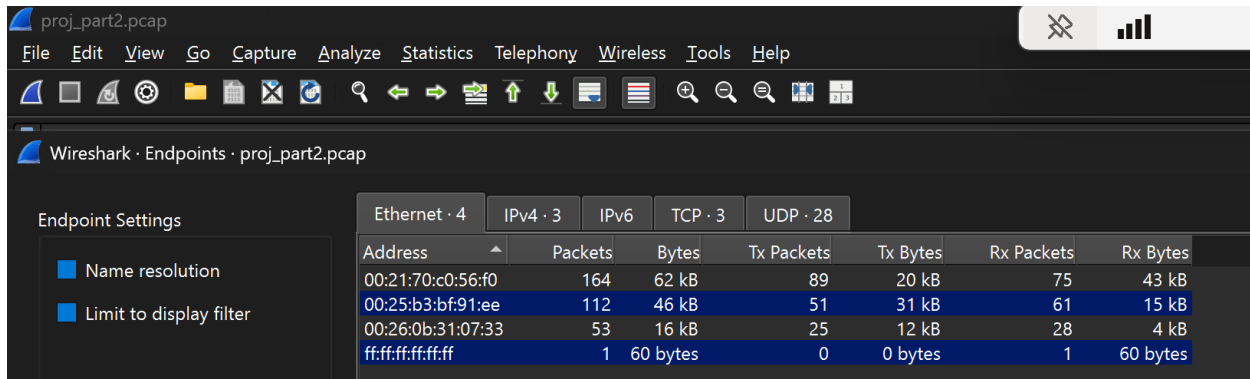
In a MITM attack, the attacker can redirect traffic between two hosts to intercept data or modify data. The ARP cache can be poisoned such that two legitimate hosts are made to think they are communicating with each other when they are actually communicating with a third party, the attacker. The attacker gets in the middle of the two hosts and relays messages between them.

1. Located the provided *proj_part2.pcap* file and opened the .pcap file in Wireshark and verified the file hash using *Statistics > Capture File Properties*.



We need to verify file hash before starting the investigation because verifying the file hash ensures the integrity and authenticity of the evidence. Also, it confirms that the packet capture file has not been altered or tampered while we do the investigation.

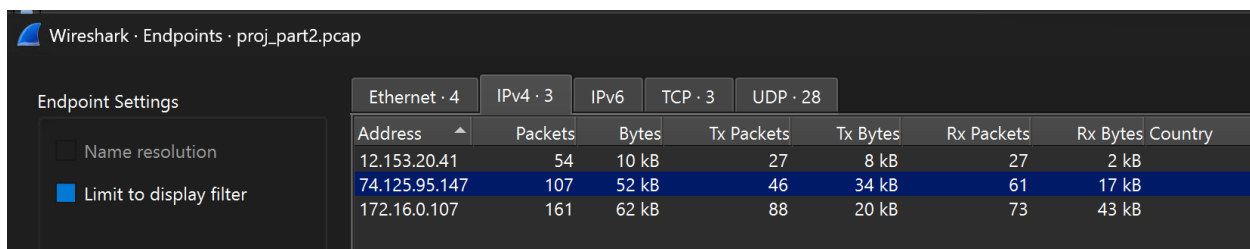
2. Navigated to the menu, Statistics > Endpoint to examine MAC, IP address and protocol information.



The screenshot shows the Wireshark interface with the 'Statistics' menu open and the 'Endpoint' sub-menu selected. The 'Ethernet' tab is active in the Endpoint Statistics pane. The table displays communication statistics for various MAC addresses.

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
00:21:70:c0:56:f0	164	62 kB	89	20 kB	75	43 kB
00:25:b3:bf:91:ee	112	46 kB	51	31 kB	61	15 kB
00:26:0b:31:07:33	53	16 kB	25	12 kB	28	4 kB
ff:ff:ff:ff:ff:ff	1	60 bytes	0	0 bytes	1	60 bytes

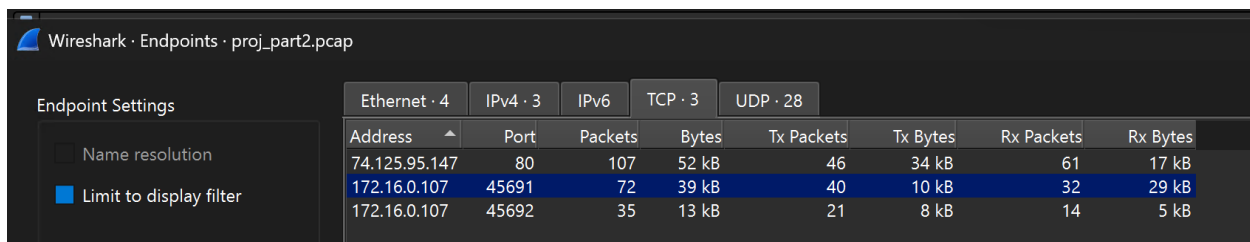
The *Ethernet* tab displayed all MAC addresses of devices involved in communication. Victim (00:21:70:c0:56:f0) and router (00:26:0b:31:07:33) MAC addresses were observed with the attacker's MAC address (00:25:b3:bf:91:ee). Also, I observed the broadcast MAC address (ff:ff:ff:ff:ff:ff).



The screenshot shows the Wireshark interface with the 'Statistics' menu open and the 'Endpoint' sub-menu selected. The 'IPv4' tab is active in the Endpoint Statistics pane. The table displays communication statistics for various IP addresses.

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country
12.153.20.41	54	10 kB	27	8 kB	27	2 kB	
74.125.95.147	107	52 kB	46	34 kB	61	17 kB	
172.16.0.107	161	62 kB	88	20 kB	73	43 kB	

The *IPv4* tab displayed three IPv4 addresses, including victim (172.16.0.107), 12.153.20.41, and 74.125.95.147.



The screenshot shows the Wireshark interface with the 'Statistics' menu open and the 'Endpoint' sub-menu selected. The 'TCP' tab is active in the Endpoint Statistics pane. The table displays communication statistics for various IP addresses and ports.

Address	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
74.125.95.147	80	107	52 kB	46	34 kB	61	17 kB
172.16.0.107	45691	72	39 kB	40	10 kB	32	29 kB
172.16.0.107	45692	35	13 kB	21	8 kB	14	5 kB

The *TCP* tab displayed communication between the victim (172.16.0.107) and an external IP address (74.125.95.147) over port 80. Also, multiple packets and bytes indicate ongoing data exchange which may be web traffic activity.

Wireshark - Endpoints · proj_part2.pcap

Endpoint Settings

Name resolution
☒ Limit to display filter

Copy
 Map

Protocol

- ☐ Bluetooth
- ☐ BPPv7
- ☐ DCCP
- ☒ Ethernet
- ☐ FC
- ☐ FDDI
- ☐ IEEE 802.11
- ☒ IEEE 802.15.4
- ☒ IPv4
- ☒ IPv6
- ☐ IPX
- ☐ JXTA
- ☐ LTP

Ethernet · 4	Port	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
12.153.20.41	53	54	10 kB	27	8 kB	27	2 kB
172.16.0.107	33189	2	248 bytes	1	78 bytes	1	170 bytes
172.16.0.107	33435	2	263 bytes	1	78 bytes	1	185 bytes
172.16.0.107	33714	2	405 bytes	1	76 bytes	1	329 bytes
172.16.0.107	33920	2	384 bytes	1	75 bytes	1	309 bytes
172.16.0.107	34580	2	333 bytes	1	79 bytes	1	254 bytes
172.16.0.107	36758	2	209 bytes	1	74 bytes	1	135 bytes
172.16.0.107	39607	2	409 bytes	1	77 bytes	1	332 bytes
172.16.0.107	40177	2	413 bytes	1	80 bytes	1	333 bytes
172.16.0.107	40919	2	207 bytes	1	78 bytes	1	129 bytes
172.16.0.107	42422	2	546 bytes	1	72 bytes	1	474 bytes
172.16.0.107	42460	2	368 bytes	1	90 bytes	1	278 bytes
172.16.0.107	42580	2	377 bytes	1	75 bytes	1	302 bytes
172.16.0.107	43010	2	406 bytes	1	75 bytes	1	331 bytes
172.16.0.107	43729	2	412 bytes	1	78 bytes	1	334 bytes
172.16.0.107	44815	2	406 bytes	1	76 bytes	1	330 bytes
172.16.0.107	45143	2	593 bytes	1	76 bytes	1	517 bytes
172.16.0.107	47755	2	364 bytes	1	79 bytes	1	285 bytes
172.16.0.107	49632	2	240 bytes	1	85 bytes	1	155 bytes
172.16.0.107	53709	2	386 bytes	1	80 bytes	1	306 bytes
172.16.0.107	54062	2	506 bytes	1	73 bytes	1	433 bytes
172.16.0.107	56676	2	354 bytes	1	85 bytes	1	269 bytes
172.16.0.107	56735	2	415 bytes	1	81 bytes	1	334 bytes
172.16.0.107	57652	2	403 bytes	1	75 bytes	1	328 bytes
172.16.0.107	57692	2	400 bytes	1	74 bytes	1	326 bytes
172.16.0.107	58402	2	324 bytes	1	76 bytes	1	248 bytes
172.16.0.107	58609	2	403 bytes	1	75 bytes	1	328 bytes
172.16.0.107	59318	2	405 bytes	1	76 bytes	1	329 bytes

The *UDP* tab displayed multiple short communications primarily involving the victim (172.16.0.107) using various ephemeral ports. Also, communication with port 53 (DNS) was observed, indicating that DNS queries were being performed by the victim.

3. Created a new column named *Source Mac* for Source MAC address using *Edit > Preferences > Columns*. Selected *Hw src addr (resolved)* in the drop-down list as Type. Then it was placed under *Source* column.

Wireshark - Preferences

Appearance

- Columns
- Font and Colors
- Layout
- Capture
- Expert
- Filter Buttons
- Name Resolution
- Protocols
- RSA Keys
- Statistics
- Advanced

Displayed	Title	Type	Fields	Field Occurrence	Resolved	Width	Alignment
☒	No.	Number			-1		Default
☒	Time	Time (format as specified)			-1		Default
☒	Source	Source address			-1		Default
☒	Source Mac	Hw src addr (resolved)			-1		Default
☒	Destination	Destination address			-1		Default
☒	Dest Mac	Hw dest addr (resolved)			-1		Default
☒	Protocol	Protocol			-1		Default
☒	Length	Packet length (bytes)			-1		Default
☒	Info	Information			-1		Default

4. Created a new column named *Dest Mac* for Source MAC address using *Edit > Preferences > Columns*. Selected *Hw dest addr (resolved)* in the drop-down list as Type. Then it was placed under *Destination* column.

Wireshark - Preferences

Appearance

- Columns
- Font and Colors
- Layout
- Capture
- Expert
- Filter Buttons
- Name Resolution
- Protocols
- RSA Keys
- Statistics
- Advanced

Displayed	Title	Type	Fields	Field Occurrence	Resolved	Width	Alignment
☒	No.	Number			-1		Default
☒	Time	Time (format as specified)			-1		Default
☒	Source	Source address			-1		Default
☒	Source Mac	Hw src addr (resolved)			-1		Default
☒	Destination	Destination address			-1		Default
☒	Dest Mac	Hw dest addr (resolved)			-1		Default
☒	Protocol	Protocol			-1		Default
☒	Length	Packet length (bytes)			-1		Default
☒	Info	Information			-1		Default

5. Clicked Apply > OK to implement the settings and captured the main pane showing the newly added columns.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length Info
1	0.000000	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	74 Standard query 0x18cd A www.google.com
2	0.001829	172.16.0.107	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	DNS	326 Standard query response 0x18cd A www.google.com CNAME www.1.google.com A 74.125.95...
3	0.018656	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	74 45691 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=586595 TSecr=0 WS=64
4	0.034180	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	74 80 → 45691 [SYN, ACK] Seq=0 Ack=1 Win=5672 Len=0 MSS=1380 TSval=2834267783 TSecr=58...
5	0.034218	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45691 → 80 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=586595 TSecr=2834267783
6	0.034283	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	HTTP	684 GET / HTTP/1.1
7	0.057644	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	66 80 → 45691 [ACK] Seq=1 Ack=619 Win=6912 Len=0 TSval=2834267806 TSecr=586595
8	0.076239	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=1 Ack=619 Win=6912 Len=1368 TSval=2834267824 TSecr=586595 [TC...
9	0.076265	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45691 → 80 [ACK] Seq=619 Ack=1369 Win=8768 Len=0 TSval=586606 TSecr=2834267824
10	0.076367	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=1369 Ack=619 Win=6912 Len=1368 TSval=2834267824 TSecr=586595 ...
11	0.076375	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45691 → 80 [ACK] Seq=619 Ack=2737 Win=11648 Len=0 TSval=586606 TSecr=2834267824
12	0.076427	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	1184 80 → 45691 [PSH, ACK] Seq=2737 Ack=619 Win=6912 Len=1118 TSval=2834267824 TSecr=58...
13	0.076439	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45691 → 80 [ACK] Seq=619 Ack=3855 Win=14400 Len=0 TSval=586606 TSecr=2834267824
14	0.076580	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=3855 Ack=619 Win=6912 Len=1368 TSval=2834267824 TSecr=586595 ...
15	0.076587	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45691 → 80 [ACK] Seq=619 Ack=5223 Win=17280 Len=0 TSval=586606 TSecr=2834267824
16	0.099663	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	HTTP	1099 HTTP/1.1 200 OK (text/html)
17	0.099714	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45691 → 80 [ACK] Seq=619 Ack=6256 Win=20032 Len=0 TSval=586612 TSecr=2834267848
18	0.205601	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	HTTP	917 GET /csi/v=3&webhp&action=&=17259,18168,24483,25233,25460,25475,25511,25529,255...
19	0.208465	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	76 Standard query 0xb2a A video.google.com
20	0.217289	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	74 45692 → 80 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM TSval=586641 TSecr=0 WS=64
21	0.232916	12.153.20.41	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	DNS	330 Standard query response 0xb2a A video.google.com CNAME video.1.google.com A 74.12...
22	0.233149	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75 Standard query 0xb747 A maps.google.com
23	0.240224	74.125.95.147	Cisco_31:07:33	172.16.0.107	Dell_c0:56:f0	TCP	74 80 → 45692 [SYN, ACK] Seq=0 Ack=1 Win=5672 Len=0 MSS=1380 TSval=3213587172 TSecr=5...

6. Then ARP traffic was identified using Wireshark's coloring rules.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length Info
100	8.576960	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	HTTP/1.1	200 OK, JSON (application/json)
103	8.743445	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	832 80 → 45692 [PSH, ACK] Seq=3333 Ack=5566 Win=16832 Len=766 TSval=3213595674 TSecr=...
105	8.743938	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	HTTP/1.1	200 OK, JSON (application/json)
108	8.825635	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=18328 Ack=7008 Win=19968 Len=1368 TSval=2834276573 TSecr=5887...
110	8.825690	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	199 80 → 45691 [PSH, ACK] Seq=11696 Ack=7008 Win=19968 Len=133 TSval=2834276573 TSecr=5887...
112	9.079793	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=11829 Ack=7008 Win=19968 Len=1368 TSval=2834276827 TSecr=5887...
114	9.080008	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=13197 Ack=7008 Win=19968 Len=1368 TSval=2834276827 TSecr=5887...
116	9.080151	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1426 80 → 45691 [PSH, ACK] Seq=14565 Ack=7008 Win=19968 Len=1368 TSval=2834276827 TSecr=...
118	9.080300	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=15925 Ack=7008 Win=19968 Len=1368 TSval=2834276827 TSecr=5887...
120	9.080496	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=17293 Ack=7008 Win=19968 Len=1368 TSval=2834276827 TSecr=5887...
122	9.080543	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1365 80 → 45691 [PSH, ACK] Seq=18661 Ack=7008 Win=19968 Len=1299 TSval=2834276827 TSecr=...
124	9.096474	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=19960 Ack=7008 Win=19968 Len=1368 TSval=2834276839 TSecr=5887...
126	9.096583	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=21328 Ack=7008 Win=19968 Len=1368 TSval=2834276839 TSecr=5887...
128	9.096665	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1426 80 → 45691 [PSH, ACK] Seq=22696 Ack=7008 Win=19968 Len=1368 TSval=2834276839 TSecr=5887...
130	9.104027	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=24056 Ack=7008 Win=19968 Len=1368 TSval=2834276852 TSecr=5888...
132	9.104322	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	TCP	1434 80 → 45691 [ACK] Seq=25424 Ack=7008 Win=19968 Len=1368 TSval=2834276852 TSecr=5888...
134	9.104337	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	HTTP/1.1	200 OK, JSON (application/json)
137	9.252425	74.125.95.147	HewlettPacka_bf:91...	172.16.0.107	Dell_c0:56:f0	HTTP	281 HTTP/1.1 204 No Content
139	4.644441	Dell_c0:56:f0	Dell_c0:56:f0	HewlettPacka_bf:91...	HewlettPacka_bf:91...	ARP	42 172.16.0.107 is at 00:21:70:c0:56:f0
141	4.644380	HewlettPacka_bf:91...	HewlettPacka_bf:91...	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60 Who has 172.16.0.107? Tell 172.16.0.1
143	4.646355	HewlettPacka_bf:91...	HewlettPacka_bf:91...	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60 172.16.0.1 is at 00:25:b3:bf:91:ee
155	14.392550	HewlettPacka_bf:91...	HewlettPacka_bf:91...	Broadcast	Broadcast	ARP	60 Who has 172.16.0.1? Tell 172.16.0.105

7. A display filter *arp* was applied to filter and analyze only ARP packets in the capture.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length Info
55	4.646442	Dell_c0:56:f0	Dell_c0:56:f0	HewlettPacka_bf:91...	HewlettPacka_bf:91...	ARP	42 172.16.0.107 is at 00:21:70:c0:56:f0
54	4.646389	HewlettPacka_bf:91...	HewlettPacka_bf:91...	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60 Who has 172.16.0.107? Tell 172.16.0.1
56	4.646455	HewlettPacka_bf:91...	HewlettPacka_bf:91...	Dell_c0:56:f0	Dell_c0:56:f0	ARP	60 172.16.0.1 is at 00:25:b3:bf:91:ee
165	14.392550	HewlettPacka_bf:91...	HewlettPacka_bf:91...	Broadcast	Broadcast	ARP	60 Who has 172.16.0.1? Tell 172.16.0.105

8. Clicked on packet 40 and examined the source and destination addresses in the packet details pane.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length Info
40	8.250804	172.16.0.107	Dell_c0:56:f0	74.125.95.147	Cisco_31:07:33	TCP	66 45692 → 80 [ACK] Seq=806 Ack=216 Win=6912 Len=0 TSval=586661 TSecr=3213587226
42	0.339478	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75 Standard query 0x3be2 A b1ogssearch.google.com
44	0.356399	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75 Standard query 0x5743 A www.youtube.com
46	0.364946	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	80 Standard query 0x9105 A picasaweb.google.com
48	0.395961	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75 Standard query 0x1bca A docs.google.com
50	0.422701	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	76 Standard query 0x6b69 A sites.google.com
52	0.424538	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	77 Standard query 0x3be2 A groups.google.com
57	6.553250	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91...	HTTP	968 GET /complete/gsearch?hl=en&client=hp&expids=17259,18168,24483,25233,25460,25475,2...

Frame 40: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface 0

Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)

Internet Protocol Version 4, Src: 172.16.0.107, Dst: 74.125.95.147

Transmission Control Protocol, Src Port: 45692, Dst Port: 80, Seq: 806, Ack: 216, Len: 0

Communication is normal because it matches the expected mapping between the victim

and router IP. The source is Victim IPv4 172.16.0.107 (00:21:70:c0:56:f0) and the destination is 74.125.95.147 via the router MAC (00:26:0b:31:07:33).

9. Clicked on packet 57 and examined the source and destination addresses in the packet details pane.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length	Info
46	0.364946	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	88	Standard query 0x9105 A picasaweb.google.com
48	0.395961	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	75	Standard query 0xb1ca A docs.google.com
50	0.422791	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	76	Standard query 0xb6b9 A sites.google.com
52	0.424530	172.16.0.107	Dell_c0:56:f0	12.153.20.41	Cisco_31:07:33	DNS	77	Standard query 0x3be2 A groups.google.com
57	6.553250	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	HTTP	960	GET /complete/gsearch?hl=en&client=hp&expIds=17259,18168,24483,25233,25460,25475,2...
59	6.593514	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	TCP	66	45691 → 80 [ACK] Seq=2364 Ack=7189 Win=25472 Len=0 TSval=588235 TSecr=2834274341
60	6.713788	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	HTTP	1005	GET /complete/gsearch?hl=en&client=hp&expIds=17259,18168,24483,25233,25460,25475,2...
62	6.743231	172.16.0.107	Dell_c0:56:f0	74.125.95.147	HewlettPacka_bf:91:ee	TCP	66	45691 → 80 [ACK] Seq=2364 Ack=7209 Win=25472 Len=0 TSval=588273 TSecr=2834274491

Frame 57: 960 bytes on wire (7680 bits), 960 bytes captured (7680 bits) on interface 0

Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: HewlettPacka_bf:91:ee (00:25:b3:bf:91:ee)

Internet Protocol Version 4, Src: 172.16.0.107, Dst: 74.125.95.147

Transmission Control Protocol, Src Port: 45691, Dst Port: 80, Seq: 1470, Ack: 6471, Len: 894

Hypertext Transfer Protocol

Starting from packet 57, abnormal behavior is observed. The victim IPv4 address remains as Source IP and destination MAC address changes to the attacker's MAC address (00:25:b3:bf:91:ee). This clearly indicates that traffic is being redirected. This is a clear indication of ARP poisoning is occurring, where the attacker is impersonating another device on the network.

10. Clicked on packet 165 and the ARP request details were examined in the packet details pane.

No.	Time	Source	Source Mac	Destination	Dest Mac	Protocol	Length	Info
139	9.104027	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	TCP	1434	80 → 45691 [ACK] Seq=24056 Ack=7908 Win=19968 Len=1368 TSval=2834276852 TSecr=588...
132	9.104332	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	TCP	1434	80 → 45691 [ACK] Seq=25424 Ack=7908 Win=19968 Len=1368 TSval=2834276852 TSecr=588...
134	9.104337	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	HTTP/2...	318	HTTP/1.1 200 OK, JSON (application/json)
137	9.292425	74.125.95.147	HewlettPacka_bf:91:ee	172.16.0.107	Dell_c0:56:f0	HTTP	281	HTTP/1.1 204 No Content
154	9.404442	Dell_c0:56:f0	Dell_c0:56:f0	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	ARP	42	172.16.0.107 is at 00:21:70:c0:56:f0
154	9.404590	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	Dell_c0:56:f0	Dell_c0:56:f0	ARP	42	Who has 172.16.0.107? Tell 172.16.0.107
154	9.404655	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	Dell_c0:56:f0	Dell_c0:56:f0	ARP	42	172.16.0.107 is at 00:21:70:c0:56:f0
165	14.392559	HewlettPacka_bf:91:ee	HewlettPacka_bf:91:ee	Broadcast	Broadcast	ARP	60	Who has 172.16.0.107? Tell 172.16.0.105

Frame 165: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

Ethernet II, Src: HewlettPacka_bf:91:ee (00:25:b3:bf:91:ee), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

Packet 165 indicates that an ARP request is being broadcast by the attacker MAC address (00:25:b3:bf:91:ee) to the network. Also, the attacker is asking “Who has 172.16.0.107?”, which indicates that it is actively participating in ARP communication. Further, “Tell 172.16.0.105” tells to send the reply to the device with IP address 172.16.0.105.

Conclusion

This project work confirms the presence of a Man-in-the-Middle (MITM) attack using ARP poisoning. The investigation started with verification of the evidence through hash validation, ensuring the integrity and reliability of the data analyzed. Then the network traffic examined established a baseline of normal communication, where the victim device communicated with external systems through the router using the correct MAC address.

Starting from packet 57, abnormal behavior is observed due to destination MAC address changes to the attacker's MAC address (00:25:b3:bf:91:ee). This is a clear indication of ARP

poisoning is occurring, where the attacker is impersonating another device on the network. Packet 165 indicates that an ARP request is being broadcast by the attacker MAC address (00:25:b3:bf:91:ee) to the network. This is demonstrated persistence, a key indicator of an ongoing attack rather than an isolated event.

These findings clearly indicate that a MITM attack occurred and highlights the importance of network monitoring and forensic analysis in detecting and responding to security incidents. Sametime, the requirement for implementing protective measures against ARP-based attacks.

References

1. Sheward, M. (2018). *Hands-on incident response and digital forensics*. BCS, The Chartered Institute for IT
2. Kinzie, K. (2022, August 19). *How to use Wireshark: Comprehensive tutorial + tips*. Varonis. <https://www.varonis.com/blog/how-to-use-wireshark>
3. Imperva. (n.d.). *ARP spoofing*. <https://www.imperva.com/learn/application-security/arp-spoofing/>
4. Chandru, J., & Bagyalakshmi, C. (2025). *MITM attack-based detection and prevention for ARP poisoning in wireless network environment*. IEEE. <https://ieeexplore.ieee.org/document/11020359>
5. Invicti. (n.d.). *ARP spoofing*. <https://www.invicti.com/learn/mitm-arp-spoofing>
6. Minnesota State University. (n.d.). Minnesota State Learning Management System.